

ZIMFEAST

Food Delivery Platform Taste the Extraordinary

APP DESIGN & DEVELOPMENT SPECIFICATION Version 1.0 | Confidential

Tishanyq Digital Harare, Zimbabwe

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1. Project Overview

1.1 Platform Summary

Field	Detail
Platform Name	ZimFeast
Tagline	Taste the Extraordinary
Business Type	Three-sided food delivery marketplace (Customers, Restaurants, Drivers)
Target Market	Zimbabwe — nationwide rollout, starting with Harare
Client / Owner	Tishanyq Digital, Harare, Zimbabwe
Currency	USD (primary), ZiG (Zimbabwe Gold) support planned
Payment Gateway	Paynow — Zimbabwe-specific (EcoCash, OneMoney, InnBucks)
Platform Type	Web App (React SPA) + Android Customer App + Android Driver App + Restaurant Dashboard + Admin Panel

1.2 Platform Description

ZimFeast is a food delivery marketplace that connects three primary user types within the Zimbabwean market:

- **Customers** browse restaurant menus, place orders, select delivery or pickup, pay via Paynow-integrated mobile money (EcoCash, OneMoney, InnBucks) or vouchers, and track their delivery in real-time on a live

map.

- **Restaurants** receive and manage incoming orders through a real-time dashboard with sound alerts, maintain their menu and pricing, view financial summaries, and print receipts for kitchen and customer records.
- **Drivers** accept delivery jobs pushed via Socket.IO, navigate to pickup and drop-off locations with in-app turn-by-turn guidance, update delivery status at each stage, capture proof-of-delivery photos, and track their daily earnings.

The platform is built on a microservices architecture consisting of six backend services: three Django services (auth, restaurant, payment) and three Go services (order, driver, realtime). All services are containerized with Docker and orchestrated via Docker Compose behind an Nginx API gateway. Real-time communication is handled through Socket.IO and Redis pub/sub. Google Maps powers location services, geocoding, and navigation across all client applications.

1.3 Additional User Types

User Type	Description
Admin	Full platform management — user/restaurant/driver/order oversight, financial analytics, promo management, driver approval workflow, banner campaigns, system health monitoring
Corporate Admin	Multi-employee company accounts with centralized billing, employee spending limits, and corporate order management

1.4 Architecture Summary

Layer	Technology
Frontend (Web)	React 18, TypeScript, Vite, Tailwind CSS, Radix UI / shadcn, TanStack Query
Frontend (Mobile)	Android Native — Java, XML layouts, Material Design 3
Backend (Django)	3 Django 4.2 microservices: auth-service (8001), restaurant-service (8002), payment-service (8005)
Backend (Go)	3 Go microservices: order-service (8003), driver-service (8004), realtime-service (3001)
Database	PostgreSQL — 1 database per service (full isolation)
Cache / PubSub	Redis 7
Authentication	Stateless JWT — shared secret, independently validated by each service
API Gateway	Nginx — routes <code>/api/*</code> to correct microservice
Maps & Navigation	Google Maps SDK (Android), Google Maps JavaScript API (Web), Directions API, Places API, Geocoding API
Payments	Paynow (Zimbabwe) — EcoCash, OneMoney, InnBucks
Real-time	Socket.IO via realtime-service (Go), Redis pub/sub for inter-service event propagation
Containers	Docker + Docker Compose with auto-scaling support
Infrastructure	AWS (Terraform-managed)

1.5 Service Communication

Pattern	Mechanism	Used By
Synchronous REST	HTTP calls with X-Service-Key header	Django services via <code>shared/service_client.py</code> , Go services via <code>go-shared/</code>
Asynchronous Events	Redis pub/sub channels	All services — order state changes, driver location updates, payment confirmations
Real-time Push	Socket.IO WebSocket connections	Customer order tracking, driver delivery offers, restaurant new-order alerts
Authentication	Stateless JWT Bearer tokens	All client-facing endpoints — each service validates independently
Data References	Cross-service UUID references (no foreign keys)	Orders reference <code>user_id</code> , <code>restaurant_id</code> , <code>driver_id</code> from other services

2. Brand Identity & Design Guidelines

2.1 Color Palette

The ZimFeast color system is built around a warm, energetic orange palette that conveys appetite appeal, speed, and premium quality. All colors are defined in the Android `colors.xml` and Tailwind CSS configuration for cross-platform consistency.

Color	Hex Code	CSS Variable	Usage
Primary Orange	#F97316	--primary	Main CTA buttons, highlights, active states, gradient starts, navigation accents
Dark Orange	#EA580C	--primary-dark	Hover states, pressed states, gradient endpoints, button hover
Deep Orange	#C2410C	--accent-deep	Strong accents, gradient variants, emphasis elements
Secondary Orange	#FB923C	--secondary	Secondary accents, lighter highlights, inactive tab indicators
Light Orange	#FDBA74	--accent-light	Subtle backgrounds, secondary variants, soft badges
Background	#F9FAFB	--background	Main screen backgrounds, page containers
Surface White	#FFFFFF	--surface	Cards, modals, panels, dialog backgrounds
Dark Text	#1F2937	--text-primary	Headings, body text, primary content
Secondary Text	#6B7280	--text-secondary	Subtext, captions, timestamps, metadata
Tertiary Text	#9CA3AF	--text-tertiary	Placeholders, hint text, disabled labels

Divider	#E5E7EB	--border	Card borders, section dividers, input borders
Error / Alert	#EF4444	--destructive	Error messages, cancellation badges, form validation, delete actions
Success	#22C55E	--success	Order confirmed, delivered status, online indicators, completion states
Warning	#F59E0B	--warning	Pending states, attention badges, expiring timers
Rating Star	#FBBF24	--rating	Star ratings across all interfaces
Link Blue	#2563EB	--link	Clickable links, navigation text, informational callouts

2.2 Typography

Category	Font Family	Weights	Usage
Primary	Plus Jakarta Sans	200 (ExtraLight) through 800 (ExtraBold)	All headings, body text, buttons, labels — primary typeface across web and marketing
Secondary	Inter, DM Sans	400–700	Alternative body text, data tables, form inputs
Monospace	JetBrains Mono, Fira Code	400–600	Order IDs, financial figures, code displays, receipt formatting
Mobile System	System sans-serif	Material Design 3 type scale	Android native app text — inherits device font with MD3 sizing

Type Scale (Web)

Element	Font	Size	Weight	Line Height
Page Title (H1)	Plus Jakarta Sans	36px	800	1.2
Section Title (H2)	Plus Jakarta Sans	28px	700	1.3
Card Title (H3)	Plus Jakarta Sans	20px	600	1.4
Body	Plus Jakarta Sans	16px	400	1.5
Caption	Plus Jakarta Sans	14px	400	1.5
Small / Badge	Plus Jakarta Sans	12px	500	1.4

Mobile Type Scale: Follows Material Design 3 type scale definitions with `sp` units. All text sizes defined in `dimens.xml` for consistency.

2.3 Icons & Imagery

Platform	Icon System	Details
Web	Lucide React	Outlined, consistent stroke weight, used for all navigation and UI icons

Web (supplementary)	Font Awesome 6.5	Brand icons, social media, extended icon set
Android	Material Design 3 Icons	System-consistent, vector drawables (XML), supports tinting
Android (custom)	Vector Drawables	Custom icons (e.g., <code>ic_my_location.xml</code>) as resolution-independent SVG-based XML

Photography Style

- Food imagery: Bright, high-contrast, warm tones, appetizing close-ups
- Restaurant photos: Full-width hero banners with overlay gradients for text legibility
- Loaded via Glide (Android) with placeholder shimmer effects

UI Visual Effects

- Glass morphism: Translucent panels with backdrop blur for depth and modernity
- Orange gradients: Linear gradients from Primary Orange to Dark Orange on headers, CTAs, and hero sections
- Rounded corners: 24dp on cards (Android), `rounded-2xl` on web cards
- Shadows: Subtle elevation shadows on cards and floating action buttons
- Animations: Float/hover animations, shimmer loading skeletons, pulse effects on live status indicators

2.4 Overall UI Tone

The ZimFeast interface prioritizes clarity, warmth, and premium feel:

Principle	Implementation
Clean & Premium	Orange palette feels warm and energetic without being cheap — generous whitespace, consistent spacing
Glass Morphism	Translucent overlay panels with blur for depth, used on navigation overlays and modal backgrounds
Dark Mode	Full dark mode support — web (Tailwind <code>class</code> strategy toggle), Android (night resource qualifiers)
Mobile-First Responsive	All web layouts designed mobile-first with <code>sm</code> : / <code>md</code> : / <code>lg</code> : Tailwind breakpoints
Material Design 3	Android apps follow MD3 guidelines with custom orange theme, dynamic color where supported
Minimal Clutter	Each screen shows only what is needed — progressive disclosure for secondary actions
Consistent Feedback	Toast notifications, loading spinners, skeleton screens, and sound alerts for state changes

3. User Types & Apps

ZimFeast serves its marketplace through five separate interfaces, each tailored to the specific workflow of its user type.

3.1 Customer App (Android Native — Java)

Attribute	Detail
Platform	Android (minSdk 24+)
Language	Java
UI Framework	XML layouts, Material Design 3, ConstraintLayout
Architecture	Activity-based with Retrofit (REST), Room (local DB), Socket.IO (real-time)
Key Libraries	Retrofit 2, Room, Glide, Google Maps SDK, Google Places API, Socket.IO Client, Biometric API
Local Storage	Room database (cart items, saved addresses), EncryptedSharedPreferences (auth tokens, biometric flag)

Screen Count: 15 screens/activities

Screen	Activity Class	Purpose
Splash	SplashActivity	JWT validation, biometric prompt, auto-routing
Landing	LandingActivity	First launch — cuisine carousel, curated restaurants, feature highlights
Register	RegisterActivity	Account creation with role selection and referral code
Login	LoginActivity	Email/password authentication, biometric login on return visits
Home	CustomerActivity	Restaurant browsing — filters, search, grids, top rated, nearby
Menu	MenuActivity	Restaurant detail — cover image, menu items, add to cart
Cart	CartActivity	Cart management — quantities, delivery/pickup toggle, address, scheduling, tips
Address Picker	AddressPickerActivity	Google Places autocomplete, draggable map pin, reverse geocoding
Address Book	AddressBookActivity	Saved addresses CRUD
Checkout	CheckoutActivity	Payment method selection, promo codes, referral credits, voucher balance
PayNow WebView	PayNowWebViewActivity	Embedded Paynow gateway (EcoCash, OneMoney, InnBucks)
Order Tracking	OrderTrackingActivity	Live Google Map with driver tracking, status timeline, driver info, ETA
Order History	OrderHistoryActivity	Past orders list with status badges

Referral	ReferralActivity	Referral code display, share link, credits tracking
Settings	SettingsActivity	Biometric toggle, WhatsApp support link

Key Capabilities:

- Room database-backed cart with offline persistence
- Socket.IO live order tracking with Google Maps markers (blue=driver, green=restaurant, red=delivery)
- Biometric authentication via Android BiometricPrompt API with EncryptedSharedPreferences
- Paynow payment integration supporting EcoCash, OneMoney, InnBucks via WebView and mobile flows
- Google Places autocomplete bounded to Zimbabwe for address entry
- Distance-based delivery fee calculation
- Promo code validation and referral credit system (15% discount, ZF-XXXXXX format codes)
- Scheduled orders with Material date and time pickers

3.2 Customer Web App (React SPA)

Attribute	Detail
Framework	React 18 + TypeScript
Build Tool	Vite
Styling	Tailwind CSS + shadcn/ui (Radix primitives)
State Management	TanStack Query (server state), React hooks (local state)
Routing	Wouter (lightweight client-side router)
Key Path	/customer (role-guarded)

Components & Features:

Feature	Component(s)	Details
Restaurant Browsing	RestaurantGrid, AllRestaurants, TopRestaurants	Paginated grid, top-rated carousel, nearby sorting by geolocation
Cuisine Filtering	QuickFilters	Horizontal filter chips: All, Fast Food, Traditional, Breakfast, Pizza, Chinese, Indian, Lunch Pack
Search	Header (integrated search)	Real-time text search across restaurant names
Menu & Cart	MenuDialog, CartComponent	Modal menu overlay, quantity controls, running total
Chef Zim AI	ChefZimCard, ChefZimDialog, ChefZimResults	AI-powered food recommendations — conversational dialog with personalized suggestions
Banner Promotions	BannerCarousel	Auto-rotating promotional banners from admin-managed campaigns
Referral System	ReferralCard	Referral code display with social sharing (WhatsApp, copy link)
Order Tracking	OrderTracking, OrderTrackingButton	Live order status with map, floating tracking button for active orders

Rating	RatingDialog	Post-delivery rating dialog — separate ratings for restaurant and driver (1-5 stars + comment)
Checkout	Checkout page	Full checkout flow with Paynow integration
Payment Return	PaymentReturn page	Handles Paynow callback/redirect after payment
Offline Support	OfflineBanner	Network status detection with banner notification
WhatsApp Support	Navbar integration	Opens wa.me/263781603382 for direct customer support

3.3 Driver App (Android Native — Java)

Attribute	Detail
Platform	Android (minSdk 24+)
Language	Java
UI Framework	XML layouts, Material Design 3, ConstraintLayout
Architecture	Activity-based with Retrofit, Socket.IO, Foreground Service
Key Libraries	Retrofit 2, Google Maps SDK, Google Directions API, Socket.IO Client, FusedLocationProvider, CameraX
Background Services	LocationService (foreground), DeliveryNotificationService

Screen Count: 4 main screens + 2 background services

Screen / Service	Class	Purpose
Login	LoginActivity	Driver authentication, role validation
Home	MainActivity	Delivery offers, daily finances, online toggle, bottom navigation (Home/Earnings/Settings)
Active Delivery	DeliveryActivity	In-app map navigation, status progression, proof of delivery, customer contact
Order History	OrderHistoryActivity	Completed deliveries list
Location Service	LocationService	Foreground service — adaptive GPS intervals, Socket.IO location emission
Notification Service	DeliveryNotificationService	Push notifications for new delivery offers

Key Capabilities:

- Real-time delivery offers via Socket.IO with 30-second countdown timer
- In-app Google Maps with route polyline, ETA/distance overlay, and re-center button
- `NavigationHelper` class manages map rendering, route drawing, and directions API calls

- 5-phase delivery status progression with server sync and Socket.IO emission
- Proof of delivery photo capture via device camera with server upload
- Background location tracking as a foreground service (adaptive: 10s active / 60s idle)
- Daily finance dashboard (earnings, deliveries, tips, hours online, average rating)
- Bottom navigation with three tabs: Home, Earnings, Settings
- Edit profile dialog, WhatsApp support, connection status indicator
- Recent reviews display on earnings tab
- Network monitoring with offline Snackbar alerts

3.4 Restaurant Dashboard (React Web)

Attribute	Detail
Framework	React 18 + TypeScript
Key Path	/restaurant (role-guarded)
Layout	DashboardLayout with tabbed interface
Real-time	WebSocket connection for live order updates

Tabs & Components:

Tab	Components	Features
Dashboard	DashboardHeader, StatsCards, LiveOrders	Restaurant open/close toggle, today's stats (orders, revenue, rating, avg order value, preparing/pending/completed counts), live order cards with status actions
Menu	MenuManagement	Full CRUD for menu items — name, description, price, image, availability toggle
Finance	RestaurantFinance	Revenue tracking, order-by-order breakdown, settlement summaries
Settings	RestaurantSettings, ExternalAPIDialog	Restaurant profile editing, external API integration for chain restaurants

Live Order Features:

Feature	Implementation
Sound Alerts	Web Audio API ascending chime (C5-E5-G5) on new orders, mute toggle persisted in localStorage
Visual Flash	New order cards flash/highlight animation for immediate attention
Status Actions	One-click status updates: Preparing, Ready, Collected
Prep Timer	PrepTimer component — countdown timer per order for kitchen staff
Receipt Printing	PrintableReceipt — opens print dialog with formatted 80mm thermal receipt (Courier New, order items, totals, delivery method)

Order Filtering	Filter by status: All, Paid, Preparing, Ready, Completed
Pagination	Cursor-based pagination for order history

3.5 Admin Panel (React Web)

Attribute	Detail
Framework	React 18 + TypeScript
Key Path	/admin/* (role-guarded, admin only)
Layout	AdminLayout — collapsible sidebar with 13 navigation items, responsive mobile drawer
Charts	Recharts library for analytics visualizations

Admin Pages:

Page	Route	Key Features
Dashboard	/admin/dashboard	Platform overview — real-time KPIs, order volume charts, revenue trends, active users
Orders	/admin/orders	All orders list with filters, search, status badges. Detail view at /admin/orders/:id
Finance	/admin/finance	Platform-wide financial analytics — revenue, fees, settlements, payment method breakdown
Users	/admin/users	User management — list, search, filter by role, view/edit profiles. Detail at /admin/users/:id
Restaurants	/admin/restaurants	Restaurant management — approval status, performance metrics. Detail at /admin/restaurants/:id
Drivers	/admin/drivers	Driver management — status, ratings, earnings history. Detail at /admin/drivers/:id
Approvals	/admin/drivers/pending	Pending driver approval workflow — review license, vehicle details, ID/vehicle photos, approve/reject
Banners	/admin/banners	Promotional banner campaign management — create, schedule, target audience, image upload
Corporate	/admin/corporate	Corporate account management — company profiles, employee spending limits, billing
Promos	/admin/promos	Promo code management — create codes, set discount type/value, expiry, usage limits
Reviews	/admin/reviews	Platform-wide review moderation — filter, flag, respond
System	/admin/system	System health monitoring — service status, error rates, performance metrics

Settings	/admin/settings	Admin account settings, platform configuration
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Admin Sidebar Navigation:

Icon	Label	Route
LayoutDashboard	Dashboard	/admin/dashboard
ShoppingBag	Orders	/admin/orders
DollarSign	Finance	/admin/finance
Users	Users	/admin/users
Store	Restaurants	/admin/restaurants
Truck	Drivers	/admin/drivers
UserCheck	Approvals	/admin/drivers/pending
Image	Banners	/admin/banners
Building2	Corporate	/admin/corporate
Tag	Promos	/admin/promos
Star	Reviews	/admin/reviews
Activity	System	/admin/system
Settings	Settings	/admin/settings

Admin Capabilities:

- CSV/Excel data export via `ExportButton` component and `exportUtils.ts`
- Real-time analytics dashboards with Recharts (line, bar, area, pie charts)
- Role-based access control — only `admin` role can access `/admin/*` routes
- Responsive design — collapsible sidebar on desktop, slide-out drawer on mobile
- Global search across all entities
- Legacy admin routes maintained for backward compatibility (`/zimfeast/admin/*`)

4. Customer App — Screen Specifications

4.1 Onboarding & Authentication

4.1.1 Splash Screen (`SplashActivity`)

Element	Details	Notes
Layout	ZimFeast logo centered on branded background	<code>activity_splash.xml</code>
Duration	1500ms display before authentication check	<code>Handler.postDelayed</code>

Token Validation	Checks <code>TokenManager.isLoggedIn()</code> and <code>isTokenValid()</code>	JWT expiry validation
Biometric Check	If biometric enabled and device supports it, shows <code>BiometricPrompt</code>	<code>BiometricHelper.showBiometricPrompt()</code>
Backend Verification	Calls <code>GET /api/auth/profile/</code> to validate token with server	Falls back to cached token on network error
Routing Logic	Valid token -> <code>CustomerActivity</code> / Invalid/expired -> <code>LandingActivity</code>	Clears tokens on 401/403 response
Offline Handling	If network fails but local token is valid, proceeds to home screen	Graceful degradation

4.1.2 Landing Screen (`LandingActivity`)

Element	Details	Notes
Auto-scroll Cuisine Carousel	10 cuisine categories with food imagery (Burger, Pizza, Sushi, Tacos, Salad, Chicken, Pasta, Dessert, Coffee, Indian)	Unsplash images, duplicated for infinite scroll effect, auto-scrolling with <code>Handler</code>
Curated Restaurant Showcase	5 premium restaurants: Neon Umami Sushi (4.8), Obsidian Burger Lab (4.9), Lush Greenery Bowl (4.6), Inferno Pizza Co. (4.7), Spice Route Delhi (4.9)	Horizontal scrolling <code>CuratedSpotsAdapter</code> cards
Feature Highlights	Speed, Security, Premium quality feature cards with images	Loaded via Glide from Unsplash
Search Container	Tappable search bar — redirects to Login (requires auth)	Visual prompt to engage
Get Started Button (top)	Routes to <code>RegisterActivity</code>	Primary CTA
Login Button	Routes to <code>LoginActivity</code>	Secondary CTA
Get Started Button (bottom)	Routes to <code>RegisterActivity</code>	Repeated CTA at page end
Auth Check	If already logged in (<code>TokenManager.isLoggedIn()</code>), auto-redirects to <code>CustomerActivity</code>	Skip landing for returning users

4.1.3 Sign Up (`RegisterActivity`)

Field	Type	Validation	Notes
First Name	<code>EditText</code>	Required, non-empty	<code>et_first_name</code>
Last Name	<code>EditText</code>	Required, non-empty	<code>et_last_name</code>
Email	<code>EditText</code>	Required, non-empty	<code>et_email</code>

Phone Number	EditText	Optional	et_phone — Zimbabwe format
Password	EditText	Required, non-empty	et_password
Role Selector	Spinner	Customer / Restaurant / Driver	Maps to customer, restaurant, driver API values
Referral Code	EditText	Optional, auto-uppercased	et_referral_code — ZF-xxxxxx format
Register Button	Button	Submits to POST /api/auth/register/	Shows loading state during request
Login Link	TextView	Navigates back to LoginActivity	tv_login — "Already have an account?"
Post-Registration	Auto-login, saves JWT tokens via TokenManager	Prompts biometric setup via BiometricHelper if device supports it	

4.1.4 Login (LoginActivity)

Element	Details	Notes
Email	EditText input	Required
Password	EditText input (password masked)	Required
Login Button	Calls POST /api/auth/login/, saves JWT access + refresh tokens	TokenManager stores in EncryptedSharedPreferences
Biometric Login	On subsequent visits, SplashActivity handles biometric before backend call	Uses Android BiometricPrompt API
Error Handling	Toast messages for invalid credentials, network errors	Graceful failure

4.1.5 Location Permission

Element	Details	Notes
System Dialog	Requests ACCESS_FINE_LOCATION permission	Android runtime permission
Purpose	Required for: delivery fee calculation (distance-based), nearby restaurant sorting, current location in address picker	FusedLocationProviderClient
Fallback	App functions without location — nearby section hidden, manual address entry required	Non-blocking permission

4.2 Home Screen (CustomerActivity)

Element	Details	Notes
Toolbar	Custom toolbar with hamburger menu icon	<code>setSupportActionBar</code> , no title
Navigation Drawer	Slide-out <code>DrawerLayout</code> with <code>NavigationView</code>	Contains: user info header, Orders, Favorites, Referrals, Help, Settings, Logout
Search Bar	<code>EditText</code> with <code>TextWatcher</code> for real-time filtering	Filters restaurants by name across all sections
Cuisine Filter Row	Horizontal chip/button row: All, Fast Food, Traditional, Breakfast, Pizza, Chinese, Indian, Lunch Pack	<code>selectedCuisine</code> state filters restaurant list
All Restaurants Grid	<code>RecyclerView</code> with <code>GridLayoutManager(2)</code> — paginated	<code>RestaurantAdapter</code> — shows: name, image (Glide), rating, cuisine type, distance
Top Rated Section	Horizontal <code>RecyclerView</code> with <code>LinearLayoutManager(HORIZONTAL)</code>	Filtered by rating ≥ 4.5 , premium showcase cards
Nearby Section	Horizontal <code>RecyclerView</code> , sorted by distance to user	Requires location permission; hidden if unavailable
Cart Badge	Floating cart button with item count badge	Backed by Room database <code>CartItem</code> table, <code>observeCart()</code> <code>LiveData</code>
Currency Toggle	USD / ZIG currency selector	<code>currentCurrency</code> state, affects displayed prices
Chef Zim AI Card	Promotional card linking to AI recommendation dialog	Web only — not present in Android MVP
Restaurant Tap	Opens <code>MenuActivity</code> with restaurant ID	<code>OnRestaurantClickListener</code> interface

4.3 Restaurant / Menu Screen (`MenuActivity`)

Element	Details	Notes
Restaurant Header	Full-width cover image, restaurant name, cuisine type, star rating, estimated delivery time	Hero section with gradient overlay
Menu Items List	Vertical <code>RecyclerView</code> — item name, description, price (USD), image, "Add to Cart" button	<code>MenuAdapter</code> , fetched from <code>GET /api/restaurants/{id}/menu-data/</code>
Add to Cart	Taps insert into Room <code>CartItem</code> table with <code>restaurantId</code> , <code>itemId</code> , name, price, quantity	Multi-restaurant cart clearing prompt if switching restaurants
Floating Cart Button	Anchored at bottom — shows total items count, tappable to open <code>CartActivity</code>	Badge updates via Room <code>LiveData</code>

Back Navigation	Toolbar back arrow returns to <code>CustomerActivity</code>	Standard Android back handling
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4.4 Cart & Checkout

4.4.1 Cart Screen (`CartActivity`)

Element	Details	Notes
Items List	<code>RecyclerView</code> with <code>CartAdapter</code> — item name, price, quantity controls (+/-), delete button	<code>onCartItemListener</code> callbacks update Room database
Order Type Toggle	Delivery / Pickup toggle	<code>isDelivery</code> flag — hides address and delivery fee for pickup
Delivery Address	Manual text entry + "Choose on Map" button (<code>AddressPickerActivity</code>) + "Saved Addresses" button (<code>AddressBookActivity</code>)	Google Places autocomplete in picker, bounded to Zimbabwe
Delivery Fee	Calculated based on distance from restaurant to delivery address	<code>DeliveryUtils</code> helper class, shown as line item
Tip Input	Numeric input for driver tip amount	Added to order total
Schedule Order	Toggle to enable scheduling + <code>Material DatePicker</code> + <code>TimePicker</code>	<code>isScheduled</code> flag, <code>scheduledCalendar</code> stores selected date/time
Subtotal / Total	Running calculation: subtotal + delivery fee + tip	Updates on any cart or fee change
Place Order Button	Creates order via API, passes <code>orderId</code> to <code>CheckoutActivity</code>	Validates address, cart not empty
Location Detection	Auto-requests location via <code>FusedLocationProviderClient</code> , up to 3 retry attempts	Used for delivery fee pre-calculation

4.4.2 Checkout (`CheckoutActivity`)

Element	Details	Notes
Order Summary	Items list with prices, subtotal, delivery fee, tip, discount, total	Read-only summary from order API
Payment Method	Radio group: PayNow Web / PayNow Mobile / Voucher	<code>selectedPaymentMethod</code> state
PayNow Web	Redirects to embedded <code>PayNowWebViewActivity</code> with Paynow gateway	EcoCash, OneMoney, InnBucks selection in gateway
PayNow Mobile	Provider spinner (EcoCash, OneMoney, InnBucks) + phone number input	Direct mobile money push notification to user's phone

Voucher	Voucher code entry + balance check	<code>loadVoucherBalance()</code> validates remaining credit
Voucher Top-up	Checkbox to apply voucher balance against order total	Partial voucher usage supported
Promo Code	Text input + "Apply" button	<code>PromoValidation</code> response — discount type (percentage/fixed), value, expiry check
Referral Credits	Checkbox to apply referral credits	<code>loadReferralCredits()</code> — 15% discount when credits available
Direct Payment Detection	Detects if payment completed without redirect (mobile push)	<code>isDirectPayment</code> flag adjusts UI flow
Pay Button	Initiates selected payment flow	Loading state, error handling with Toast
Post-Payment	Navigates to <code>OrderTrackingActivity</code> with <code>orderId</code>	On successful payment confirmation

4.4.3 PayNow WebView (`PayNowWebViewActivity`)

Element	Details	Notes
WebView	Full-screen embedded Paynow payment gateway	JavaScript enabled, DOM storage enabled
Supported Methods	EcoCash, OneMoney, InnBucks	Zimbabwe mobile money providers
Success Callback	Detects Paynow success URL, returns result to <code>CheckoutActivity</code>	<code>WebViewClient</code> URL interception
Cancellation	Detects cancellation URL, returns cancelled result	User can press back to cancel
Error Handling	Network error page, retry option	Graceful WebView error handling

4.5 Live Order Tracking (`OrderTrackingActivity`)

Element	Details	Notes
Google Map	Embedded <code>MapView</code> with three markers	Blue (driver, real-time), Green (restaurant/origin), Red (delivery address/destination)
Route Polyline	Drawn from restaurant to delivery address	<code>PolylineOptions</code> connecting origin to destination
Map Bounds	Auto-fits camera to show all markers	<code>LatLngBounds.Builder</code> with padding
Status Timeline	Visual step indicator: Confirmed -> Preparing -> Out for Delivery -> Delivered	Updates from API polling and <code>Socket.IO</code> events
Socket.IO Connection	<code>TrackingSocketManager</code> — listens for <code>driver:location</code> and <code>order:status</code>	Real-time driver position updates

	events	
Polling Fallback	15-second base interval with exponential backoff (max 2 minutes)	consecutiveFailures counter increases interval on errors
Driver Info Card	Driver name, phone number, vehicle details	Shown when order status is "assigned" or "out_for_delivery"
ETA Display	Minutes remaining + distance from driver to destination	Updated via Socket.IO + polling
Call Driver	Button opens phone dialer with driver's number	ACTION_DIAL intent
Map Visibility	Map card shown only when driver is assigned (status = "assigned" or "out_for_delivery")	Hidden during preparing/ready phases
Rating Dialog	Auto-triggers when status transitions to "delivered"	RatingDialog — separate ratings for restaurant (1-5 stars) and driver (1-5 stars) + optional comment. Shown once per session (ratingDialogShown flag)
Back Navigation	Returns to CustomerActivity	Standard back press

4.6 Orders, Addresses & Profile

4.6.1 Order History (OrderHistoryActivity)

Element	Details	Notes
Orders List	RecyclerView with OrderHistoryAdapter	Shows: order date, status badge (color-coded), item summary, total
Status Badges	Color-coded by status: pending (yellow), preparing (blue), delivered (green), cancelled (red)	Material chip styling
Tap Action	Opens OrderTrackingActivity for order detail/status	Re-uses tracking screen for historical view
Pagination	Paginated API response	Scroll to load more

4.6.2 Address Book (AddressBookActivity)

Element	Details	Notes
Saved Addresses List	RecyclerView with AddressBookAdapter	Shows: label, full address text, lat/lng stored
Add Address	Button opens AddressPickerActivity	Returns selected address
Edit / Delete	Swipe or tap actions on address items	Room database CRUD
Select for Order	Tappable address returns result to CartActivity	startActivityForResult pattern

4.6.3 Address Picker (`AddressPickerActivity`)

Element	Details	Notes
Google Places Autocomplete	Search bar with <code>AddressSuggestionAdapter</code>	Bounded to Zimbabwe coordinates for relevant results
Map with Draggable Pin	Google Map with centered, draggable marker	User can fine-tune location after search
Reverse Geocoding	Converts map pin coordinates to human-readable address	Updates address text as pin moves
Current Location Button	Gets device GPS and centers map	<code>FusedLocationProviderClient.getLastLocation()</code>
Confirm Button	Returns selected address (text, latitude, longitude) to calling activity	<code>setResult(RESULT_OK, intent)</code>

4.6.4 Referral Screen (`ReferralActivity`)

Element	Details	Notes
Referral Code Display	User's unique code in ZF-xxxxxx format	Prominently displayed, copy-to-clipboard action
Share Link	Share intent with referral URL	Android share sheet — WhatsApp, SMS, etc.
Available Credits	Number of unused referral credits	Each successful referral earns 1 credit
Used Credits	Number of credits already applied	Historical tracking
Discount Percentage	15% discount per credit used	Displayed as benefit explanation

4.6.5 Settings (`SettingsActivity`)

Element	Details	Notes
Biometric Login Toggle	Switch to enable/disable biometric authentication	Stored in <code>EncryptedSharedPreferences</code> via <code>TokenManager</code>
WhatsApp Support	Button opens WhatsApp chat with ZimFeast support	<code>wa.me/263781603382</code> — direct support line
App Version	Displays current app version	Informational

5. Driver App — Screen Specifications

5.1 Driver Onboarding

5.1.1 Login (`LoginActivity`)

Element	Details	Notes
Email	<code>EditText</code> input	Required

Password	EditText input (masked)	Required
Login Button	Calls auth API, validates <code>role == "driver"</code>	Rejects non-driver accounts
Token Storage	Saves auth token via <code>ZimFeastDriverApp.getInstance()</code>	Application-level singleton storage
Post-Login	Fetches driver profile (name, vehicle, approval status)	Stores driver name for header display
Redirect	On success -> MainActivity	Clears LoginActivity from stack

5.1.2 Profile Setup

Field	Details	Notes
License Number	Driver's license ID	Required for approval
Vehicle Make	Vehicle manufacturer	Text input
Vehicle Model	Vehicle model name	Text input
License Plate	Registration plate number	Text input
Vehicle Color	Vehicle color description	Text input
ID Photo	Government ID photo upload	Camera capture or gallery
Vehicle Photo	Vehicle exterior photo	Camera capture or gallery
Approval Status	Admin must approve before driver can go online	<code>AdminPendingDriversPage</code> handles approval workflow

5.2 Home Screen (MainActivity)

The driver home screen uses a `BottomNavigationView` with three tabs: **Home**, **Earnings**, and **Settings**. Each tab swaps content within the same activity using visibility toggles on `LinearLayout` and `ScrollView` containers.

5.2.1 Home Tab

Element	Details	Notes
Driver Name Header	Displays driver's name from <code>ZimFeastDriverApp</code> singleton	<code>tv_driver_name</code>
Header Rating	Star rating badge in header area	<code>layout_header_rating + tv_header_rating</code> , fetched from driver profile API
Online Toggle	<code>SwitchMaterial</code> — large toggle at top	Emits <code>driver:go_online</code> / <code>driver:go_offline</code> via <code>Socket.IO</code> , starts/stops <code>LocationService</code>
Status Text	"Online — Waiting for orders" / "Offline"	<code>tv_status</code> updates with toggle state

Offline Message	Informational text when offline	tv_offline_message — hidden when online
Delivery Offer Card	Real-time Socket.IO delivery offers	card_delivery_offer — hidden by default, shown on incoming offer
Offer - Restaurant	Restaurant name and address	tv_offer_restaurant, tv_offer_address
Offer - Distance	Distance to restaurant + total delivery distance	tv_offer_distance
Offer - Earnings	Delivery fee + tip breakdown	tv_offer_earnings
Offer - Countdown	30-second CountdownTimer with visual display	tv_offer_timer — OFFER_TIMEOUT_SECONDS = 30
Offer - Accept	Accepts delivery, navigates to DeliveryActivity	btn_accept — sends acceptance via Socket.IO + REST
Offer - Decline	Declines offer, returns to waiting state	btn_decline — hides offer card
Network Monitor	NetworkUtils LiveData observer	Shows Snackbar.LENGTH_INDEFINITE when offline with dismiss action
Socket Connection	SocketManager listener interface	SocketListener callbacks for offers, status updates, connection state

5.2.2 Earnings Tab

Element	Details	Notes
Total Earnings	Today's total earnings in USD	tv_total_earnings
Total Deliveries	Number of completed deliveries today	tv_total_deliveries
Total Distance	Kilometers driven today	tv_total_distance
Total Tips	Tips collected today	tv_total_tips
Hours Online	Total hours spent online today	tv_hours_online
Average Rating	Driver's average rating score	tv_average_rating
Recent Reviews	List of recent customer reviews	layout_recent_reviews container, tv_no_reviews placeholder
View History Button	Opens OrderHistoryActivity	btn_view_history

Data source: `GET /api/drivers/daily/finances/` — per-day aggregation endpoint.

5.2.3 Settings Tab

Element	Details	Notes
Driver Name	Profile display name	tv_settings_name
Phone Number	Contact number	tv_settings_phone
Vehicle Info	Vehicle make/model/plate/color	tv_settings_vehicle
Connection Indicator	Green/red dot showing Socket.IO connection status	view_connection_indicator + tv_connection_status
Edit Profile	Opens dialog for updating vehicle details	btn_edit_profile — in-activity AlertDialog with EditText fields
WhatsApp Support	Opens WhatsApp chat with ZimFeast support	btn_whatsapp_support — wa.me/263781603382
Logout	Clears auth tokens, returns to LoginActivity	btn_logout — clears ZimFeastDriverApp session

5.3 Active Delivery (DeliveryActivity)

The active delivery screen is the driver's primary workspace during a delivery. It provides in-app map navigation, status progression, customer contact, and proof-of-delivery capture.

5.3.1 Layout Structure

Element	Details	Notes
Status Header	Current delivery phase text with gradient background	tv_delivery_status
In-App Map	Embedded <code>MapView</code> managed by <code>NavigationHelper</code>	Google Maps with route polyline, driver marker (blue), destination marker (green=pickup, red=dropoff)
ETA/Distance Overlay	Floating overlay on map showing real-time ETA and distance	overlay_eta container with tv_eta and tv_distance, updated by <code>NavigationHelper</code> from Google Directions API
Re-center Button	Floating button at bottom-right of map	btn_recenter — calls <code>navigationHelper.recenterMap()</code> to fit full route in view
Pickup/Dropoff Details	Step indicator showing restaurant name (pickup) and delivery address (dropoff)	tv_restaurant and tv_dropoff
Customer Info	Customer name with call button	customerName + customerPhone — phone dial intent
Earnings Card	Delivery fee + tip display	deliveryFee and tip values from intent extras, green accent styling
Navigate Button	Opens external Google Maps for turn-by-turn directions	btn_navigate — fallback for detailed navigation

Update Status Button	Primary action button — advances delivery to next phase	btn_update_status — text changes per phase
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5.3.2 Delivery Status Progression

The delivery follows a strict 5-phase progression. Each status update is synced to the server via REST API and broadcast via Socket.IO.

Phase	Status Code	Header Text	Button Text	Map Destination	Notes
1	driver_assigned	"Navigate to Restaurant"	"Arrived at Restaurant"	Restaurant (green marker)	Initial state — driver heads to pickup
2	arrived_restaurant	"At Restaurant - Wait for pickup"	"Picked Up Order"	Restaurant (green marker)	Driver waiting for food preparation
3	picked_up	"Order Picked Up - Heading to Customer"	"Arrived at Destination"	Customer (red marker)	In transit — map switches to dropoff destination
4	arrived_destination	"At Destination - Deliver to Customer"	"Complete Delivery"	Customer (red marker)	At customer location — ready to hand off
5	delivered	"Delivery Complete!"	"Done" (disabled)	N/A	Triggers proof-of-delivery photo prompt, auto-finishes after 2 seconds

5.3.3 Navigation System

Component	Details	Notes
NavigationHelper	Dedicated class managing Google Maps rendering	Handles MapView lifecycle, marker placement, polyline drawing, camera animation
Route Polyline	Drawn from driver position to current destination	Updates as destination changes (restaurant -> customer)
Directions API	Google Directions API for ETA and distance	Real-time updates as driver location changes
Location Updates	FusedLocationProviderClient with LocationRequest	Priority.PRIORITY_HIGH_ACCURACY, updates fed to NavigationHelper and Socket.IO
External Navigation	Opens Google Maps app with destination coordinates	Fallback for drivers who prefer full turn-by-turn

5.3.4 Proof of Delivery

Element	Details	Notes
Photo Prompt	AlertDialog asking driver to take delivery photo	Triggered when advancing to "delivered" status
Camera Launch	ActivityResultContracts.TakePicture launcher	FileProvider for secure camera file URI
Camera Permission	ActivityResultContracts.RequestPermission for CAMERA	Falls back to completing without photo if denied
Photo Upload	MultipartBody.Part upload via Retrofit	okhttp3.MediaType image/jpeg, RequestBody from file
Skip Option	If photo capture cancelled or fails, delivery completes without photo	completeDeliveryStatus() called on skip
File Storage	Temporary JPEG in app's external files directory	SimpleDateFormat timestamp naming: delivery_yyyyMMdd_HHmss.jpg

5.3.5 Location Service Integration

Action	Intent	Effect
Delivery Accepted	ACTION_DELIVERY_ACTIVE sent to LocationService	Switches to 10-second GPS interval for frequent tracking
Delivery Completed	ACTION_DELIVERY_IDLE sent to LocationService	Switches to 60-second GPS interval for battery conservation
Status Updates	Socket.IO emission	order:status_update event with orderId and new status
Location Broadcast	Socket.IO emission from LocationService	driver:location event with lat/lng for customer tracking

5.4 Background Services

5.4.1 Location Service (LocationService)

Attribute	Details
Type	Android Foreground Service (Service) with persistent notification
Notification Channel	ZimFeastDriverLocation — low importance, always visible while active
Location Provider	FusedLocationProviderClient (Google Play Services)
Active Interval	10,000ms (10 seconds) — during active delivery
Idle Interval	60,000ms (60 seconds) — when online but no active delivery
Interval Switching	ACTION_DELIVERY_ACTIVE / ACTION_DELIVERY_IDLE intents from DeliveryActivity
Location Emission	Each location update emitted via SocketManager as driver:location event

Socket Listener	Implements <code>SocketManager.SocketListener</code> — receives delivery offers while running
Delivery Notifications	Delegates to <code>DeliveryNotificationService</code> for offer push notifications
Lifecycle	<code>START_STICKY</code> — restarts if killed by system
Notification	Shows "ZimFeast Driver — Tracking your location" with tap-to-open <code>MainActivity</code>

5.4.2 Delivery Notification Service (`DeliveryNotificationService`)

Attribute	Details
Purpose	Shows Android push notifications for incoming delivery offers
Trigger	Called from <code>LocationService</code> when <code>Socket.IO</code> delivers a new offer event
Notification Content	Restaurant name, estimated earnings, distance
Tap Action	Opens <code>MainActivity</code> to display full offer card with accept/decline
Channel	Separate high-importance notification channel for delivery alerts

5.5 Earnings & History

5.5.1 Daily Finance (Home Screen — Earnings Tab)

Metric	Source	Display
Total Deliveries	<code>GET /api/drivers/daily/finances/</code>	Integer count
Total Earnings	<code>GET /api/drivers/daily/finances/</code>	USD currency format
Total Tips	<code>GET /api/drivers/daily/finances/</code>	USD currency format
Total Distance	<code>GET /api/drivers/daily/finances/</code>	Kilometers
Hours Online	<code>GET /api/drivers/daily/finances/</code>	Hours and minutes
Average Rating	<code>GET /api/drivers/daily/finances/</code>	Star rating (X.X / 5.0)
Recent Reviews	<code>GET /api/drivers/daily/finances/</code>	Customer name, rating, comment text

5.5.2 Order History (`OrderHistoryActivity`)

Element	Details	Notes
Completed Deliveries List	<code>RecyclerView</code> with <code>OrderHistoryAdapter</code>	Shows: date, restaurant, customer area, earnings, status
Pagination	Paginated API response	Scroll-to-load-more pattern
Navigation	Accessible from Earnings tab "View History" button	<code>btn_view_history</code> in <code>MainActivity</code>

6. Restaurant Dashboard — Screen Specifications

Web-based dashboard (React), also usable on tablet. Located at /restaurant route.

6.1 Authentication

- Restaurant user logs in via web login page
- JWT token stored in localStorage
- Redirects to dashboard if already authenticated

6.2 Dashboard Layout

- DashboardHeader: Restaurant name, open/closed toggle (calls POST /api/restaurants/toggle-open/)
- Tabbed interface with 4 tabs: Dashboard, Menu, Finance, Settings

6.3 Dashboard Tab (Live Orders)

StatsCards component showing:

- Today's Orders (count)
- Today's Revenue (USD)
- Average Rating (stars)
- Menu Items Count
- Average Order Value

LiveOrders component:

- Real-time order list via WebSocket (/ws/restaurants/{id}/dashboard/)
- Polling fallback every 10 seconds
- Sound alerts for new orders (Web Audio API - ascending C5/E5/G5 chime)
- Visual flash animation (amber pulse, 3 cycles, 0.67s each)
- Mute toggle button (persisted in localStorage)
- Filter tabs: All, Pending, Preparing, Ready (with counts)
- Order cards showing: order ID, items list, total, status badge, timestamp, delivery/collection method
- Action buttons: "Start Preparing" → "Mark Ready" → "Mark Collected" (for collection orders) / "Awaiting Delivery" (for delivery orders)
- Prep timer countdown for orders being prepared
- Print receipt button per order
- Pagination (Previous/Next 10)
- Highlighted orders (WebSocket new_order events get 5s orange highlight)
- Export functionality

6.4 Menu Management Tab

MenuManagement component:

- Table view: item name, price, availability toggle, prep time, category
- Add new menu item (dialog form with image upload, multipart/form-data)
- Edit existing items
- Delete items (confirmation dialog)
- Category management

6.5 Finance Tab

RestaurantFinance component:

- Today's revenue & order count
- Weekly/monthly trend charts
- Top performing items

- Payment method breakdown (pie chart)
- Earnings history with platform fee deductions
- Unsettled amounts for direct payment restaurants
- Restaurant debt tracking

6.6 Settings Tab

RestaurantSettings component:

- Restaurant name, description, cover image
- Cuisine types selection
- Minimum order price
- Opening hours configuration (supports 24h + overnight schedules, UTC+2)
- Address / location
- Open/closed manual override toggle

6.7 External API Integration

ExternalAPIDialog component:

- Register external menu API endpoint
- Configure webhook URLs for order notifications
- API key management
- Useful for chain restaurants with central menu systems

7. Admin Panel — Screen Specifications

For internal use by ZimFeast operations team. Web-based React SPA at /admin/* routes.

7.1 Navigation

Collapsible sidebar with sections:

- Main: Dashboard, Orders, Finance
- Management: Users, Restaurants, Drivers, Pending Drivers
- Marketing: Banners, Corporate, Promos
- System: Reviews, System Health, Settings

7.2 Admin Dashboard (AdminDashboardPage)

Element	Details	Notes
Live Order Stats	Active orders by status (pie chart), orders last hour, avg delivery time	Real-time via polling
Stuck Orders Alert	Orders pending longer than threshold	Actionable warning
Revenue Trend	Daily revenue area chart (last 30 days)	Recharts visualization
Status Breakdown	Orders by status bar chart	Visual distribution
Top Restaurants	Table: restaurant name, order count, revenue	Sortable

Recent Orders	Last 20 orders with status badges	Quick overview
User Counts	Total customers, restaurants, drivers, admins	Role breakdown

7.3 Orders Management (AdminOrdersPage)

Element	Details	Notes
Search	Search by order ID	Instant filter
Status Filter	Filter by order status	Dropdown select
Orders Table	ID, restaurant, customer ID, status badge, fee, created date	Sortable columns
Pagination	10/25/50 per page options	Configurable
Order Detail	Click to view full breakdown: items, quantities, customer, driver, payment, timeline	Separate detail page
Status Override	Admin can force status change on any order	Emergency control
Export	CSV/Excel export of filtered orders	ExportButton component

7.4 User Management (AdminUsersPage)

Element	Details	Notes
Role Filter	Filter by: customer, restaurant, driver, admin	Tab/dropdown
Search	Search by email or name	Instant filter
Users Table	Email, name, role badge, active status, joined date	Paginated
Suspend User	Dialog with reason text field	Toggles is_active
Promote to Admin	Elevate user role	With confirmation
Export	CSV/Excel export	Filtered data

7.5 Restaurant Management (AdminRestaurantsPage)

Element	Details	Notes
Search	Search by restaurant name	Instant filter
Restaurant Table	Name, owner, open/closed, avg rating, reviews count, created	Sortable
Suspend/Unsuspend	Toggle restaurant active status	With confirmation
Detail View	Full restaurant profile, menu, orders, earnings	Click to expand

7.6 Driver Management (AdminDriversPage)

Element	Details	Notes
Search	Search drivers by name	Instant filter
Drivers Table	Name, phone, rating (stars), total deliveries, vehicle details	Sortable
Online Status	Real-time online/offline indicator	Color-coded
Detail View	Full driver profile, delivery history, ratings, earnings	Click to expand

7.7 Pending Driver Approvals (AdminPendingDriversPage)

Element	Details	Notes
Pending List	Drivers awaiting approval: name, license, vehicle, submitted date	Queue view
Approve	One-click approval, driver can go online	POST /api/drivers/admin/{id}/approve/
Reject	Reject with notes/reason	POST /api/drivers/admin/{id}/reject/

7.8 Financial Analytics (AdminFinancePage)

Element	Details	Notes
Revenue Summary	Total revenue, this month, this week, today	KPI cards
Revenue by Payment Method	Pie chart (Paynow/Voucher/Direct)	Visual breakdown
Daily Revenue Trend	Bar + area chart (30 days)	Recharts
Top Restaurants by Revenue	Ranked table	Sortable
Failed Payments	List of failed Paynow transactions	Actionable
Settlements	Track settlements to restaurants	Outstanding amounts
Refund Processing	Admin can process refunds (mark failed, restore voucher)	Manual override

7.9 Promotional Management

Page	Elements	Notes
Banners (AdminBannersPage)	Create/edit promotional banners with campaigns (free_delivery, discount, info, new_restaurant), targeting (all/new/returning users), scheduling, priority, image	Displayed in customer app carousel
Promo Codes (AdminPromosPage)	Create discount codes: percentage/fixed, min order amount, max uses, expiry date, active toggle	Validated at checkout

Corporate (AdminCorporatePage)	Manage corporate accounts, employee limits, spending caps	Multi-employee companies
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7.10 System & Reviews

Page	Elements	Notes
Reviews (AdminReviewsPage)	All customer reviews feed	Moderation capability
System (AdminSystemPage)	Service health checks, performance metrics	Monitoring
Settings (AdminSettingsPage)	Platform fees, delivery zones, app configuration	Global settings

8. Technical Specifications

8.1 Architecture Overview

ZimFeast uses a microservices architecture with 6 backend services behind an Nginx API gateway:

Service	Port	Language	Framework	Database	Replicas	Memory
Auth Service	8001	Python	Django 4.2 + DRF	zimfeast_auth	2	512MB
Restaurant Service	8002	Python	Django 4.2 + DRF	zimfeast_restaurants	2	512MB
Order Service	8003	Go	net/http + gorilla/mux	zimfeast_orders	3	128MB
Driver Service	8004	Go	net/http + gorilla/mux	zimfeast_drivers	2	128MB
Payment Service	8005	Python	Django 4.2 + DRF	zimfeast_payments	2	512MB
Realtime Service	3001	Go	Socket.IO	Redis only	2	256MB
API Gateway	80	-	Nginx	-	1	128MB

8.2 Platforms & Frameworks

Layer	Technology
Customer App (Mobile)	Java (Native Android), Min SDK 24, Target SDK 34
Driver App (Mobile)	Java (Native Android), Min SDK 24, Target SDK 35
Web Frontend	React 18 + TypeScript, Vite 5, Tailwind CSS 3.4, Radix UI / shadcn
Data Fetching	TanStack Query (React Query), Socket.IO client
Backend (Django)	Django 4.2, Django REST Framework, Gunicorn/Daphne

Backend (Go)	Go standard library, gorilla/mux, lib/pq
Database	PostgreSQL 16 (1 database per service, 5 total)
Cache / PubSub	Redis 7 (pub/sub for async events, streams for crash recovery)
Authentication	Stateless JWT (shared secret across all services, 14-day access, 30-day refresh)
API Gateway	Nginx (path-based routing to correct service)
Real-time	Socket.IO (driver tracking, order updates, delivery offers)
Maps	Google Maps API (Places, Directions, geocoding — frontend + mobile)
Payments	Paynow Zimbabwe (EcoCash, OneMoney, InnBucks)
Push Notifications	Firebase Cloud Messaging (planned)
Storage	AWS S3 (food photos, rider photos, delivery proof images)
CDN	AWS CloudFront (frontend assets + media)
Containerization	Docker + Docker Compose (production orchestration)
Cloud	AWS (af-south-1 Cape Town region)

8.3 Inter-Service Communication

Method	Description	Used For
REST (Django)	HTTP calls via shared/service_client.py with X-Service-Key header	Cross-service data fetching
REST (Go)	HTTP calls via go-shared/ with X-Service-Key header	Cross-service data fetching
Redis Pub/Sub	JSON events on named channels	Real-time order status, delivery creation
Redis Streams	Durable message streams with consumer groups	Crash recovery, guaranteed delivery
Socket.IO	WebSocket with 2 namespaces: /drivers, /customers	Live tracking, delivery offers, status updates
JWT Forwarding	User JWT passed between services for identity	User context in cross-service calls

8.4 Database Schema Summary

Database	Key Tables	Purpose
zimfeast_auth	CustomUser, Address, BlacklistedToken, CorporateAccount, CorporateEmployee	User identity, addresses, corporate accounts

zimfeast_restaurants	Restaurant, MenuItem, Branch, RestaurantChain, RestaurantEarning, RestaurantFinanceSummary, RestaurantDebt, RestaurantReview, Banner, RestaurantExternalAPI	Restaurant profiles, menus, finance, reviews, promotions
zimfeast_orders	orders_order, orders_orderitem	Order lifecycle, items, delivery tracking
zimfeast_drivers	drivers_driver, drivers_driverorderstatus, drivers_driverfinance, drivers_driverrating, drivers_driverreject	Driver profiles, assignments, earnings, ratings
zimfeast_payments	Payment, FeastVoucher, PromoCode, PromoUsage, ReferralCode, ReferralCredit, ReferralTracking	Payments, e-wallet, promotions, referrals

8.5 Order Status State Machine

pending_payment → paid → preparing → ready → collected → assigned → out_for_delivery → delivered
 scheduled → pending_payment

any → cancelled

8.6 Payment Methods

Method	Flow	Notes
Paynow (Web)	Redirect to Paynow payment page in WebView/browser	EcoCash, OneMoney, InnBucks
Paynow (Mobile)	USSD push to customer's phone (provider-specific)	Requires phone number + provider selection
Voucher (FeastVoucher)	Deduct from e-wallet balance	Instant, no external gateway
Voucher + Paynow	Partial voucher deduction + Paynow top-up for remainder	Hybrid payment
Direct Payment	Customer pays restaurant's own Paynow account	Platform still takes commission

8.7 Delivery Fee Calculation

- Rate: \$0.35 per kilometer (Haversine distance)
- Minimum fee: \$1.50
- Geographic validation: Must be within Zimbabwe bounds (lat -22.5 to -15.3, lng 25.2 to 33.1)

8.8 Performance Requirements

- App load time: Under 3 seconds on 4G connection
- Order placement to kitchen notification: Under 10 seconds (via Redis pub/sub + Socket.IO)
- Live tracking update interval: 10 seconds (active delivery), 60 seconds (idle)
- Socket.IO reconnection: Unlimited attempts, exponential backoff (1s → 30s), 50% jitter
- Polling fallback: 15s base with exponential backoff to 2 minutes max

- Mid-range Android support: Min SDK 24 (Android 7.0+)
- Offline handling: Network status detection, offline banner, graceful degradation

8.9 Security

Feature	Implementation
API Transport	HTTPS only (ACM certificates, TLS 1.2+)
Authentication	Stateless JWT with refresh token rotation (old tokens blacklisted)
Admin Bootstrap	ADMIN_SETUP_TOKEN required for first admin account creation
Token Storage (Mobile)	EncryptedSharedPreferences (AES256-GCM) on customer app
Biometric Auth	AndroidX BiometricPrompt (BIOMETRIC_STRONG) on customer app
Sensitive Fields	Fernet symmetric encryption for Paynow keys, API secrets
Service Auth	X-Service-Key header for inter-service communication
Rate Limiting	100 req/min per IP on auth endpoints (Go), DRF throttling (Django)
WAF	AWS WAF v2 with managed rules (Common + Known Bad Inputs) + rate limiting (2000/5min/IP)
Network Isolation	RDS + Redis only accessible within VPC, no public endpoints
GPS Privacy	Driver GPS data only active when driver is online

9. AWS Deployment Architecture

9.1 Infrastructure (Terraform-managed)

Component	AWS Service	Specification
Region	af-south-1 (Cape Town)	Closest to Zimbabwe for low latency
Networking	VPC 10.0.0.0/16	2 public + 2 private subnets, Internet Gateway
Compute	ECS Fargate	6 microservices, auto-scaling (target: 70% CPU)
Database	RDS PostgreSQL 16.4	db.t4g.micro (Phase 1), auto-scaling storage 20-100GB, 7-day backups
Cache	ElastiCache Redis 7.1	cache.t3.micro, at-rest encryption
Load Balancer	Application Load Balancer	HTTPS listener, path-based routing, health checks
CDN	CloudFront	2 distributions (frontend + media), SPA fallback, PriceClass_200 (Africa)
Storage	S3	2 buckets: frontend (static) + media (uploads), public access blocked

DNS	Route 53	Root domain + www + media subdomains
SSL	ACM	Certificates in us-east-1 (CloudFront) + af-south-1 (ALB)
Security	WAF v2	Regional (ALB) + CloudFront, managed rules + rate limiting
Registry	ECR	6 repositories with lifecycle policy (keep last 10 images)
Monitoring	CloudWatch	Log groups per service, 30-day retention
IAM	Execution + Task roles	Least privilege: ECR pull, S3 media access, CloudWatch logs

9.2 Scaling Strategy

Phase	Orders/Day	Database	Estimated Cost
Phase 1 (MVP)	0–500	db.t4g.micro (free tier)	~\$110–130/month
Phase 2	500–2,000	db.t4g.small + Multi-AZ	~\$150–250/month
Phase 3	2,000+	db.t4g.medium or Aurora Serverless v2	~\$300+/month

9.3 CI/CD Pipeline (GitHub Actions)

Workflow	Trigger	Steps
deploy-services.yml	Push to main (backend/* changes) or manual	Change detection → Docker build → ECR push (SHA + latest tags) → Django migrations (ECS RunTask) → Force new ECS deployment (rolling, zero-downtime)
deploy-frontend.yml	Push to main (webapp/* changes) or manual	npm build → S3 sync (1yr cache for assets, no-cache for index.html) → CloudFront invalidation

10. Deliverables & Current Implementation Status

10.1 What Is Built (MVP Complete)

Deliverable	Status	Technology
Customer Android App	✅ Built	Java (Native Android), 15 screens
Driver Android App	✅ Built	Java (Native Android), 4 screens + background services
Customer Web App	✅ Built	React 18 + TypeScript + Tailwind
Restaurant Dashboard (Web)	✅ Built	React 18 + TypeScript + Tailwind
Admin Panel (Web)	✅ Built	React 18 + TypeScript + Tailwind, 13+ pages
Corporate Dashboard (Web)	✅ Built	React 18 + TypeScript
Auth Microservice	✅ Built	Django 4.2 + DRF
Restaurant Microservice	✅ Built	Django 4.2 + DRF

Order Microservice	✔ Built	Go
Driver Microservice	✔ Built	Go
Payment Microservice	✔ Built	Django 4.2 + DRF
Realtime Microservice	✔ Built	Go + Socket.IO
API Gateway	✔ Built	Nginx
Docker Orchestration	✔ Built	Docker Compose (319 lines)
AWS Infrastructure	✔ Built	Terraform (12 files)
CI/CD Pipeline	✔ Built	GitHub Actions (2 workflows)
Google Maps Integration	✔ Built	Places, Directions, Geocoding
Paynow Integration	✔ Built	EcoCash, OneMoney, InnBucks
Real-time Order Tracking	✔ Built	Socket.IO + Redis pub/sub
In-App Navigation (Driver)	✔ Built	Google Directions API + MapView
Sound Alerts (Restaurant)	✔ Built	Web Audio API (C5/E5/G5 chime)
WhatsApp Support	✔ Built	wa.me deep link (+263781603382)
Biometric Authentication	✔ Built	AndroidX BiometricPrompt
Referral System	✔ Built	ZF-XXXXXX codes, 15% credit
Promo Code System	✔ Built	Percentage/fixed discounts
FeastVoucher E-Wallet	✔ Built	Balance management + payment split
Corporate Accounts	✔ Built	Multi-employee, spending limits
Scheduled Orders	✔ Built	Background scheduler (30s check)
Driver Approval Workflow	✔ Built	Admin approve/reject + notes
Proof of Delivery Photos	✔ Built	Camera capture + multipart upload
Restaurant Finance Tracking	✔ Built	Per-order earnings, commission, debts
Receipt Printing	✔ Built	Browser print API
Data Export (Admin)	✔ Built	CSV/Excel (SpreadsheetML)
Banner/Promotion System	✔ Built	Scheduled campaigns, targeting
Chef Zim AI Recommendations	✔ Built	AI-powered food suggestions (web)
Dark Mode	✔ Built	Class-based toggle (web)
External Restaurant API	✔ Built	Chain restaurant integration

10.2 Phase 2 Roadmap

Feature	Priority	Notes
iOS Customer App	High	Native Swift or cross-platform
iOS Driver App	High	Native Swift or cross-platform
SMS OTP Verification	High	Africa's Talking or Twilio
Push Notifications	High	Firebase Cloud Messaging
Loyalty/Rewards Program	Medium	Points-based system
Multi-language Support	Medium	Shona, Ndebele translations
ZiG Currency Display	Medium	Dual currency (USD + ZiG)
Advanced Analytics	Medium	Customer behavior, heatmaps
Customer Favorites/Saved	Low	Persistent restaurant bookmarks
Group Orders	Low	Shared cart feature
Driver Chat	Low	In-app messaging

11. Timeline & Priorities

Phase	Scope	Target
Phase 1 (MVP) ✔	Customer app + Driver app + Web apps + All 6 backend services + Docker + AWS infrastructure + CI/CD	Complete
Phase 2	iOS apps, SMS OTP, push notifications, loyalty program, ZiG currency	Q2 2026
Phase 3	Multi-language, advanced analytics, group orders, driver chat, scheduled deliveries v2	Q3 2026

Priority	Detail
Primary Device	Android (majority of Zimbabwe smartphone users)
Internet Conditions	Must work on standard Zimbabwean mobile data (Econet, NetOne, Telecel)
Language	English only (MVP)
Currency	USD primary, ZiG dual display planned
Region	AWS af-south-1 (Cape Town) for low-latency Zimbabwe access

12. Contact & Project Owner

Field	Detail
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Platform	ZimFeast — Food Delivery Marketplace
Company	Tishanyq Digital
Location	Harare, Zimbabwe
Support WhatsApp	+263 78 160 3382
Website	zimfeast.com

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